

Where to Intervene? Benchmarking Fairness-Aware Learning on Differentially Private Synthetic Tabular Data

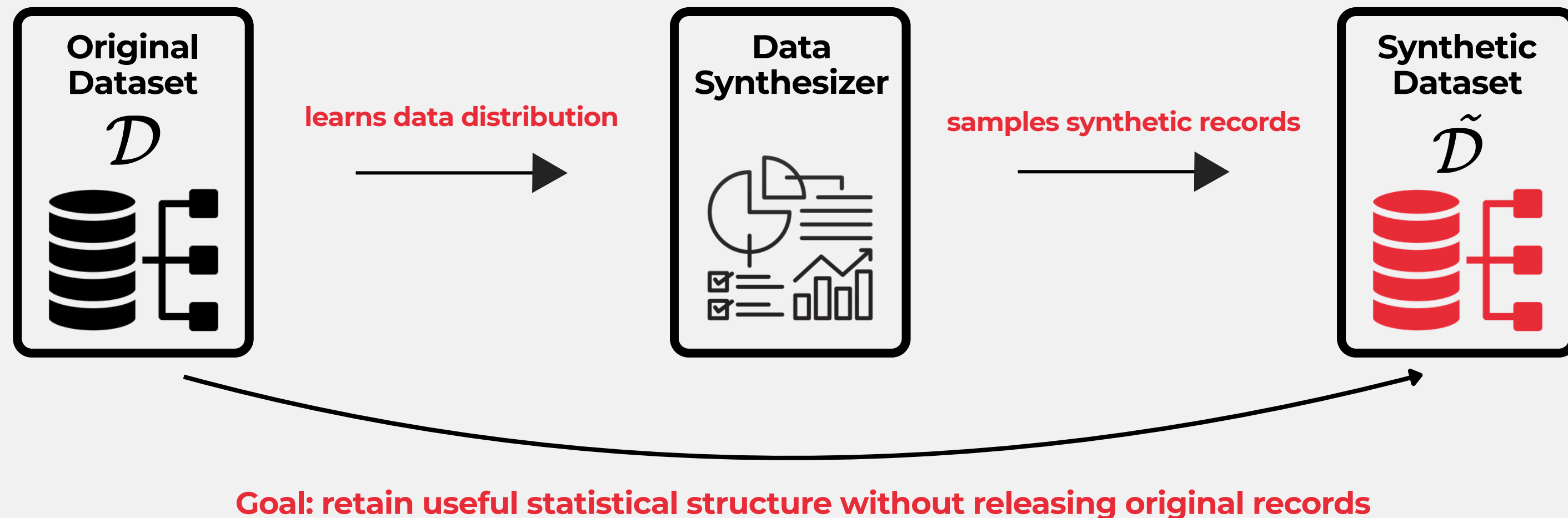
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Context & Motivation

Synthetic Data for Data Sharing

- **Adopted by industry** to share data while limiting access to original records.
- **Discussed in regulatory guidance** as a privacy-aware approach for data access and reuse [1].
- But synthetic data is **not private by default** [2]: it may still **leak** sensitive information.
- It may also **reproduce or amplify biases** already present in the original data.

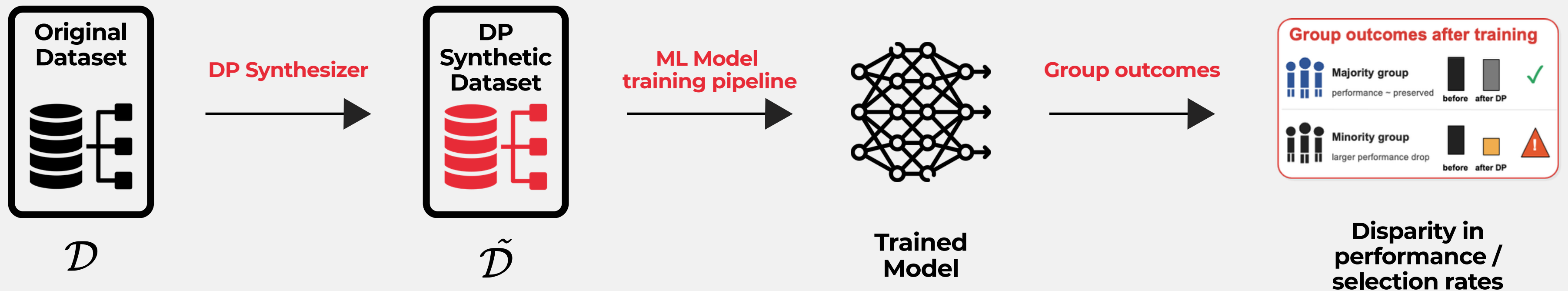


[1] FSA. 2023. Feedback Statement on Synthetic Data Call for Input. <https://www.fca.org.uk/publications/feedback-statements/fs23-1-feedback-statement-synthetic-data-call-for-input>.

[2] Georgi Ganey and Emiliano De Cristofaro. 2025. The Inadequacy of Similarity- Based Privacy Metrics: Privacy Attacks Against “Truly Anonymous” Synthetic Datasets.

Differentially Private Synthetic Data

- To address the privacy gap, one may use **Differential Privacy (DP)** [3].
- **DP synthetic data generators** [4] produce $\tilde{\mathcal{D}}$ from $\mathcal{D}$ with formal privacy guarantees.
- Downstream ML training on $\tilde{\mathcal{D}}$ is **DP post-processing**: it does not weaken the privacy guarantee.
- But DP noise may **distort subgroup structure**, affecting:
 - **Utility**: lower predictive performance;
 - **Fairness**: larger group disparities (a.k.a. "**DP disparate impact**") [5].



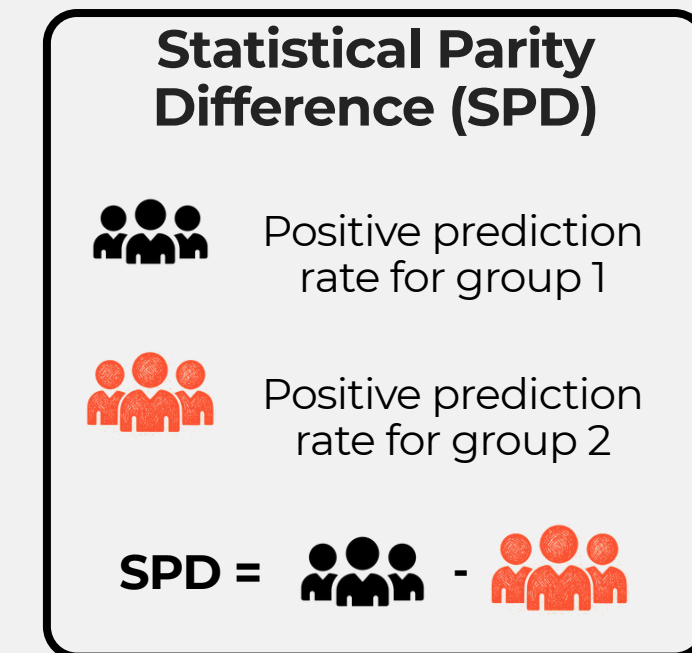
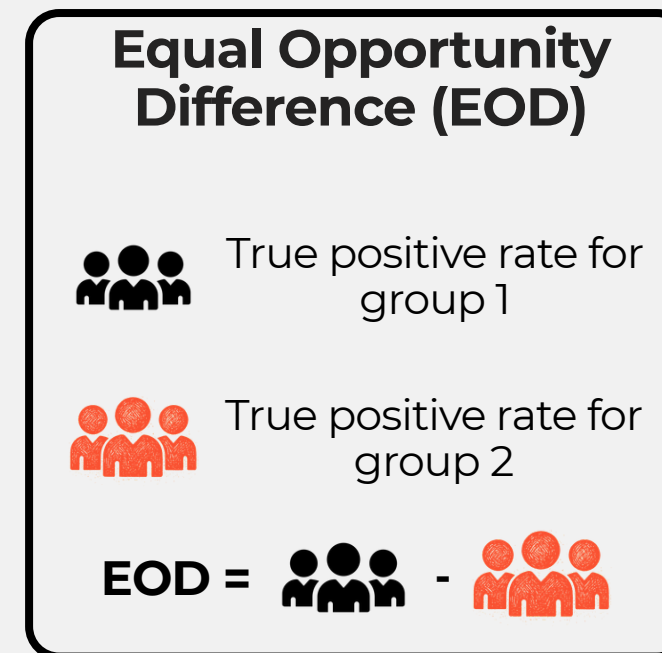
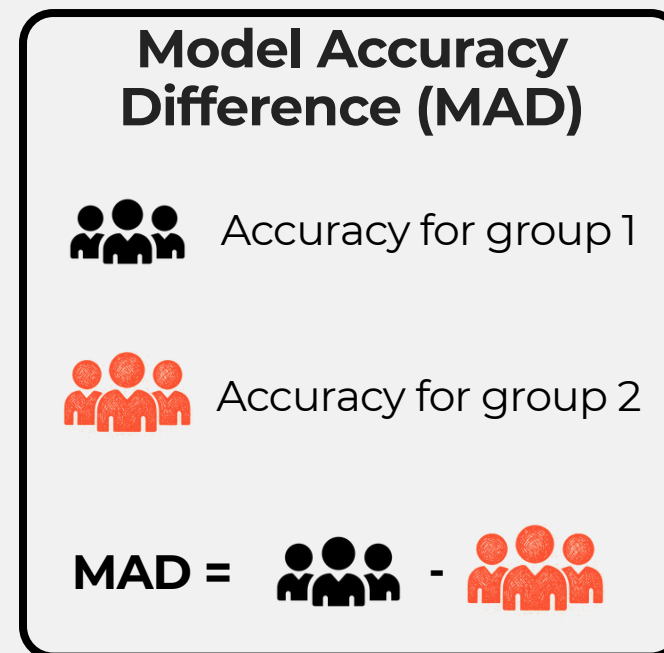
[3] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating Noise to Sensitivity in Private Data Analysis. In Theory of Cryptography.

[4] Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. 2022. AIM: an adaptive and iterative mechanism for differentially private synthetic data. VLDB 2022.

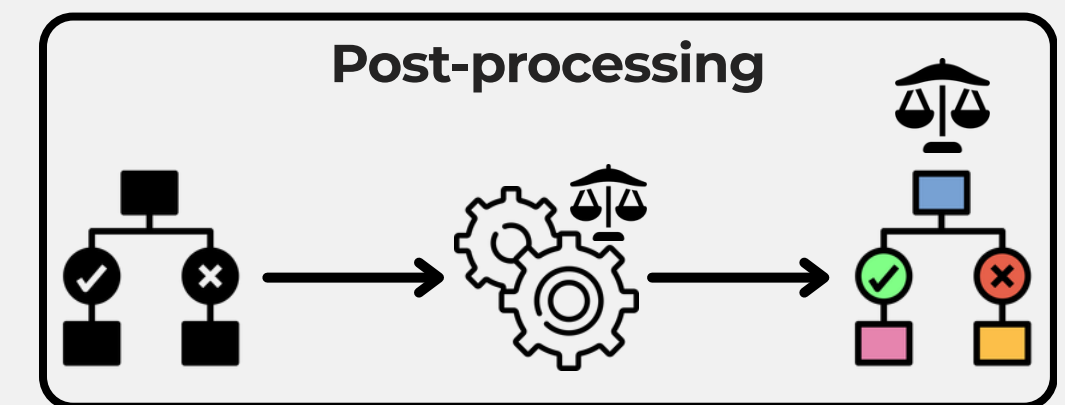
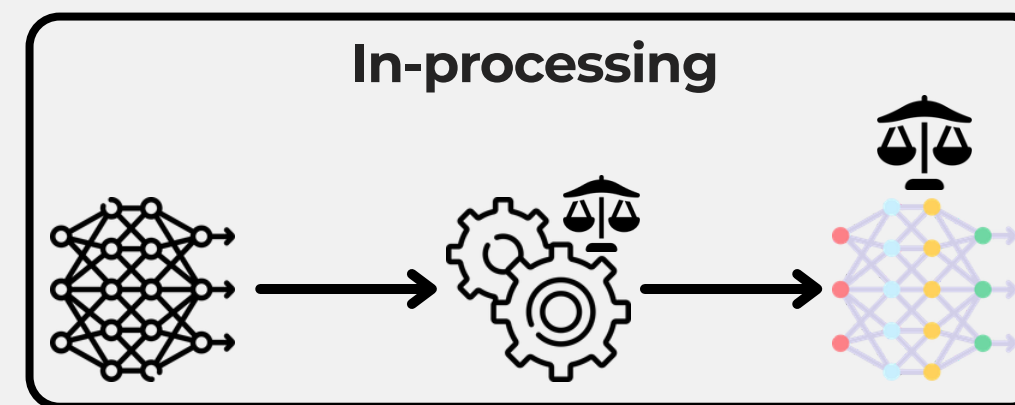
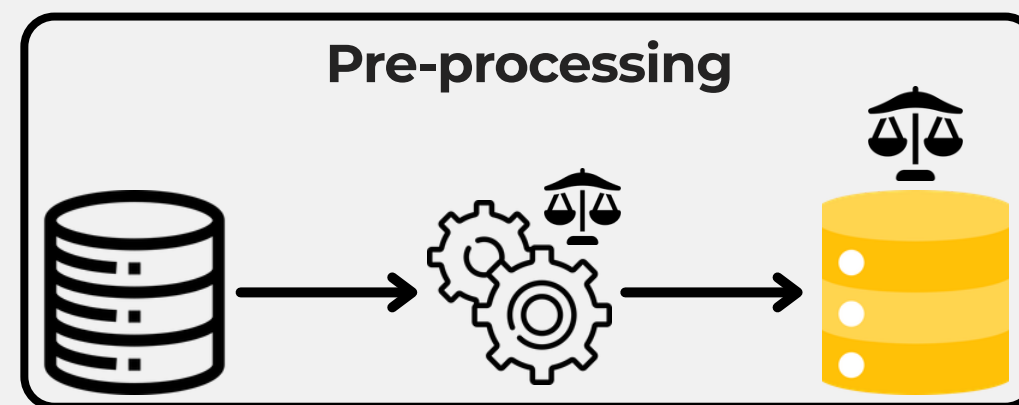
[5] Georgi Ganey, Bristena Oprisanu, and Emiliano De Cristofaro. 2022. Robin hood and Matthew effects: Differential privacy has disparate impact on synthetic data. ICML 2022.

Measuring and Mitigating Unfairness

How to measure disparity? → Group Fairness Metrics



How to correct disparity? → Fairness Mechanisms



Research Questions

Research Question 1

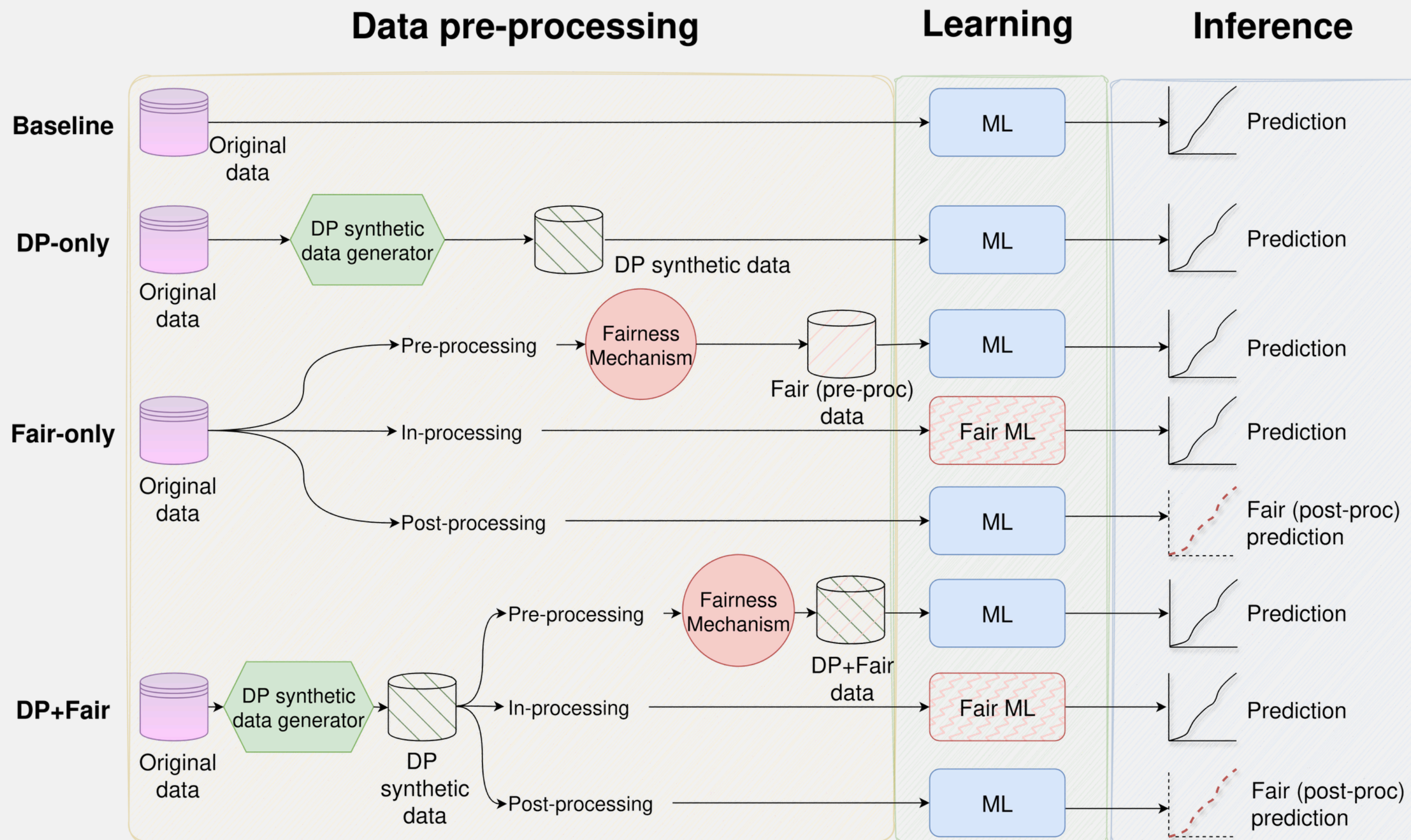
**Can fairness interventions recover
DP-induced fairness loss?**

Research Question 2

**Where should one intervene for the
best privacy-fairness-utility trade-off**

We Introduce BenchmarkDPFair

We introduce BenchmarkDPFair



Experimental Setup

Datasets

3 Real-World (Adult, ACSIncome, Compas)
BiasOnDemand: **6** controlled bias configurations

Classifiers

3 Machine learning models
(Logistic Regression, Random Forest, XGBoost)

Privacy budgets (ϵ)

12 Different values (0.05 to 20)

Seeds

20 Random seeds

DP Synthesizers

2 marginal-based synthesizers (AIM, MST)

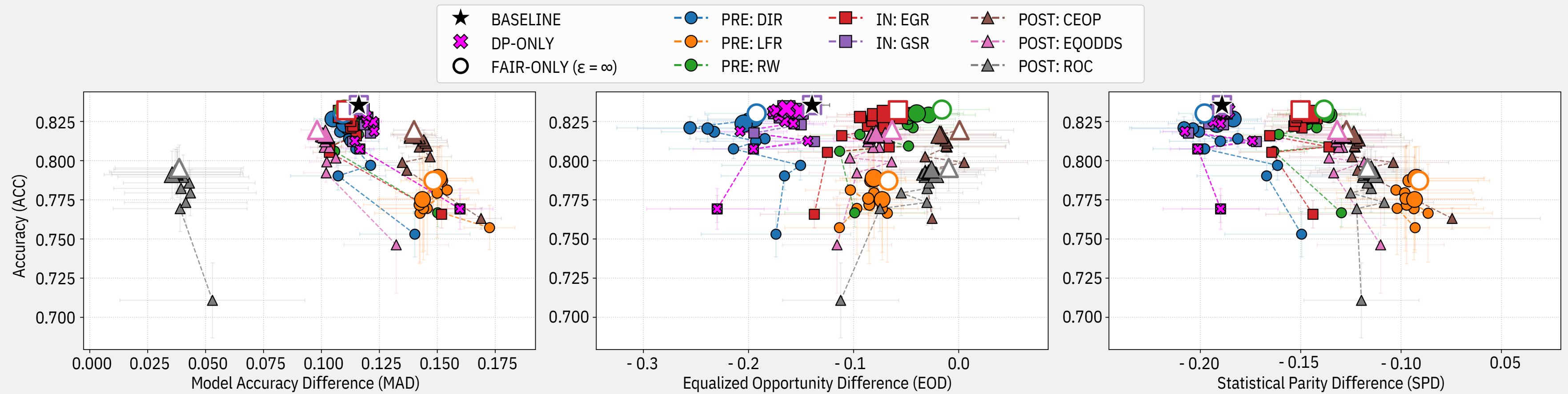
Ablation

DP-compliant calibration
In-processing hyperparameter tuning

In this talk: representative results with AIM and Adult. Full benchmark in the paper

Selected Results

DP+Fair Moves Toward Parity



Representative setting: Adult · AIM · XGBoost

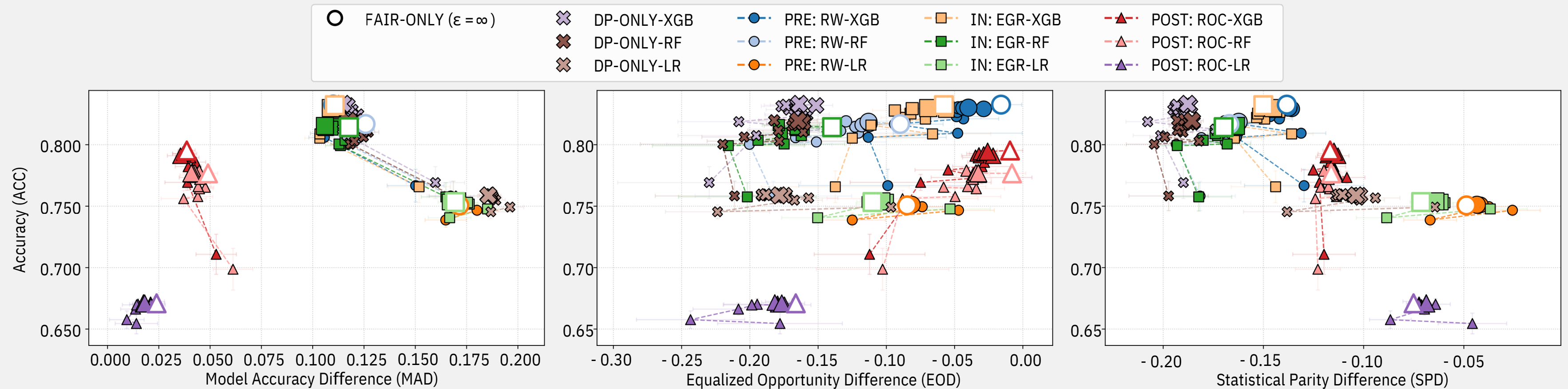
↑ **Accuracy = better utility**

→ **MAD/EOD/SPD Toward 0 = better fairness**

Larger marker = larger ϵ / weaker privacy

DP+Fair often moves closer to zero disparity than DP-only, especially for EOD and SPD

Robustness Check: Patterns Across Classifiers



Representative setting: Adult · AIM · main interventions across XGBoost, RF, and LR

↑ Accuracy = better utility

→ MAD/EOD/SPD Toward 0 = better fairness

Larger marker = larger ϵ / weaker privacy

The same qualitative trends remain visible across classifiers; the result is not an XGBoost artifact

Conclusion & Perspectives

Takeaway & Future Perspectives

Can we recover fairness?

Yes — partially
DP+Fair improves **EOD/SPD**
Effects are metric-dependent

Where should we intervene?

Post-processing gives the best
fairness-privacy-utility trade-off
ROC and **EqOdds** are the
strongest first choices

What remains open?

Beyond **binary** tabular tasks
Richer protected attributes
Joint fair-private synthesizers

BenchmarkDPFair

A reproducible and extensible testbed for privacy-fairness-utility trade-offs

pip install BenchmarkDPFair

Benchmark · Reproduce · Extend



Thank you! Questions?

Paper



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BenchmarkDPFair

